

Synthesis Conditions of $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$ and Its Structural Relatives: Evidence Grading, Route Matrix, and Transferability Limits

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Abstract

The non-centrosymmetric silicate $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$ has been grown from a high-temperature solution and its single-crystal structure solved, but direct evidence for its starting composition, its bulk phase region, and any independent reproduction is still missing. Here we grade the available evidence and compare the published compositions, thermal histories, containers, cooling programs, and characterization results for the target phase, the Zn-free tetragonal Ba–Y silicate family, structurally related Ba–Zn–Si–O compounds, and Y–Si–O process-reference systems. The existing experiments show that phase formation near the target is governed simultaneously by the Ba/Y/Si ratio, the Zn–Y–oxygen stoichiometry, the flux proportion, and the cooling path, and that single-crystal structural evidence is no substitute for bulk phase-purity and measured-composition analysis. On this basis we propose four priorities: establishing a local phase diagram around $\text{Ba}_5\text{Y}_{12}\text{ZnSi}_8\text{O}_{40}$, testing the composition–structure relationship along $\text{Ba}_5\text{Y}_{13-x}\text{Zn}_x[\text{SiO}_4]_8\text{O}_{8.5-x/2}$, running solid-state reaction and high-temperature solution growth in parallel, and using Mg/Co analogs to isolate the effect of ionic-size differences. A local two-step inorganic retrosynthesis model recommends ZnO, MgO, and Co_3O_4 , respectively, together with BaCO_3 , Y_2O_3 , and SiO_2 , as the preferred starting-material sets for the Zn, Mg, and Co targets; these are precursor hypotheses awaiting verification, nothing more. The experiments most likely to yield discovery are not further trials at isolated temperatures but a closed feedback loop that ties together composition grids, phase quantification, single-crystal structure, measured stoichiometry, and process records.

1 Introduction

This review concerns the non-centrosymmetric barium yttrium zinc silicate $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$. We pursue three questions: whether the composition and structural assignment of the target phase can be confirmed; under which phase-formation conditions the phase can be obtained reproducibly; and whether a continuous composition–structure relationship connects the Zn-containing phase to the Zn-free tetragonal Ba–Y silicates. In the original report, crystals were grown from a high-temperature solution in an open system and the structure was solved by single-crystal X-ray diffraction (SCXRD) [1]. The structure combines a framework of highly coordinated rare-earth–oxygen polyhedra with isolated or composite silicate units, and it offers a reference point both for how complex silicates assemble and for the compositional design of non-centrosymmetric oxides.

The direct report of the target phase establishes only that the crystals exist and what their structure is; it gives no independently reproducible starting composition, no phase-formation range, and no route to phase-pure bulk samples [1]. We therefore do not stitch together a “best recipe” from the conditions used for structurally related compounds. Instead, we separate the questions that single-crystal growth, solid-state re-

action, and phase-diagram screening can each answer, and we use that separation to design experiments that discriminate between alternative structural hypotheses. Literature values are quoted throughout with their stated limitations; structurally related systems and process-reference systems serve only to suggest experimental reference points.

1.1 Current Understanding of Related Systems

- **The tetragonal Ba–Y–Si–O silicate family.** Early flux-growth experiments reported $\text{Ba}_{5+x}\text{Y}_{13}\text{Si}_8\text{O}_{41}$ as an accompanying crystalline phase, and a later series of 151 runs in the $\text{BaO–K}_2\text{O–Y}_2\text{O}_3\text{–SiO}_2\text{–MoO}_3$ system showed that the $\text{Y}_2\text{O}_3/\text{SiO}_2$ ratio, the MoO_3 content, and the cooling rate together govern which silicate structure type forms and how well the crystals grow [25, 42]. Recent single-crystal work has recast the related nominal composition as non-stoichiometric $\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$, and ceramic experiments showed that grinding in agate can introduce SiO_2 and push bulk samples off the nominal ratio [43, 44]. A separate phase-region study that combined energy-dispersive spectroscopy (EDS), continuous rotation electron diffraction (cRED), synchrotron powder X-ray diffraction (PXRD), and neutron diffraction confirmed $\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$ and found competing phases and a glass-forming region [18]. Taken together, these results place the target phase inside a phase region governed jointly by the Ba/Y/Si ratio, oxygen stoichiometry, liquid composition, and thermal history, not at the end of a single isolated recipe.
- **Structurally related Ba–Zn–Si–O compounds.** BaMSiO_4 ($M = \text{Co}, \text{Zn}, \text{Mg}$) belongs to the family of stuffed-tridymite derivatives, which shows that divalent tetrahedral cations can replace one another within a given framework; it does not follow that these structures are isostructural with the Y-containing target phase [29]. The low- and high-temperature forms of $\text{BaZn}_2\text{Si}_2\text{O}_7$ can be told apart by neutron and X-ray diffraction, a reminder that the observation temperature and the cooling path both bear on structure assignment [28]. A solid-state study of $\text{BaZnSi}_3\text{O}_8$ proceeded through calcination, pelletizing, sintering, and controlled cooling, with characterization by synchrotron or laboratory X-ray diffraction [51]. The main lesson of the Ba–Zn–Si–O literature is therefore to record the Zn source, the atmosphere, the cooling path, and the phases present at both high temperature and room temperature together, rather than to borrow any single temperature outright.
- **Y–Si–O process-reference systems.** Y_2SiO_5 and $\text{Y}_2\text{Si}_2\text{O}_7$ each occur in several polymorphs and have been prepared by high-temperature solution growth, melt growth, and the Czochralski method. Diffraction, local spectroscopy, and high-temperature measurements show that each polymorph, and the product that remains after cooling, must be identified separately [6, 11, 31, 33]. These studies help standardize how crucible, flux, volatilization loss, slow cooling, and annealing conditions are recorded, but they do not show that the same growth methods apply to $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$.

1.2 Organization of the Review

We first verify the literature sources for the target phase and its structural relatives, then summarize what earlier experiments have established about composition, competing phases, contamination, and thermal history, then compare the synthesis routes and their experimental conditions, and finally propose four research directions: a local phase diagram, Zn–Y–oxygen-stoichiometry coupling, solid-state reaction run in parallel with high-temperature solution growth, and Mg/Co analogs. The two-step inorganic retrosynthesis model used here generates only a set of precursors to be tested; temperatures, atmospheres, crucibles, and characterization methods are still fixed from the literature and from materials-chemistry constraints.

Figure 1 summarizes the line of argument. Related systems are used first to establish structural relationships, competing phases, and process boundaries; phase-diagram and charge-balance analysis then proposes candidate compositions, from which the retrosynthesis model derives the precursor set; solid-state reaction and high-temperature solution growth test the bulk phase region and the single-crystal structure, respectively; and composition and process are adjusted round by round on the basis of the phase composition, the mea-

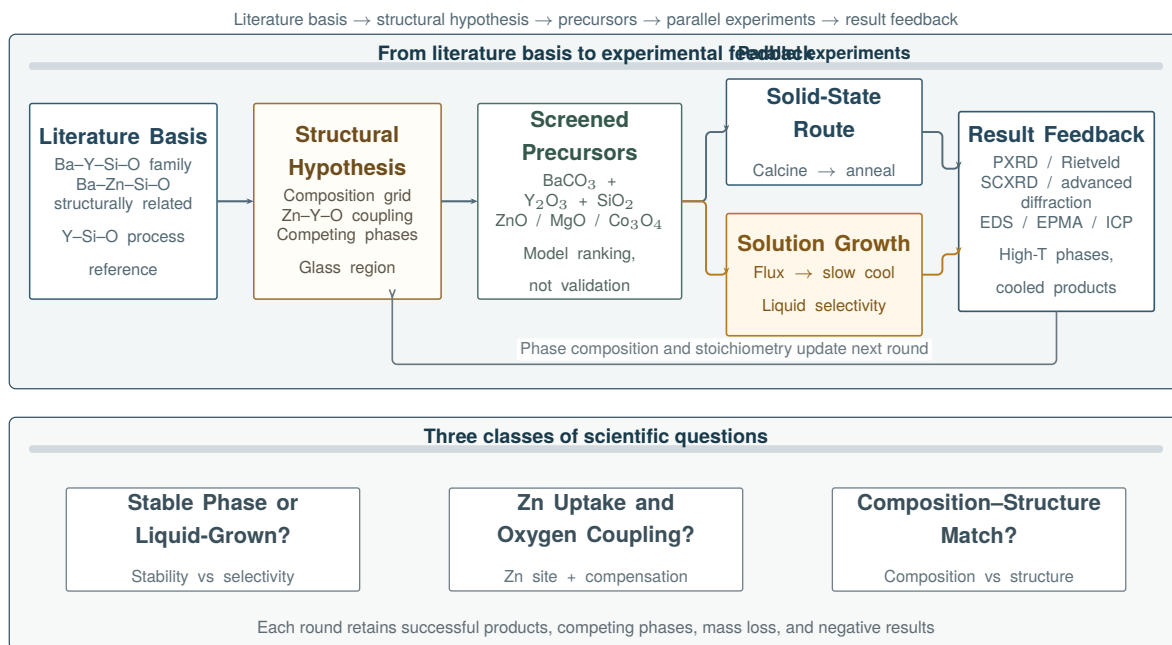


Figure 1. Research roadmap. The Ba–Y–Si–O, Ba–Zn–Si–O, and Y–Si–O systems are used first to establish structural relationships, competing phases, and process boundaries; phase-diagram and charge-balance analysis proposes candidate compositions, and the two-step inorganic retrosynthesis model supplies the precursor set; solid-state reaction and high-temperature solution growth probe the bulk phase region and the single-crystal structure, respectively, and the characterization results feed back into the next round of compositions and processes.

sured stoichiometry, and the structure refinement. Each round of experiments should eliminate at least one structural or phase-region hypothesis rather than chase an unverified “best recipe”.

2 Literature Search and Evidence Assessment

2.1 Search Scope

The search began from the target formula in both its structural and its elemental spelling, using $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$, $\text{Ba}_5\text{Y}_{12}\text{ZnSi}_8\text{O}_{40}$, BYZSO , and barium yttrium zinc silicate, combined with the keywords space group, crystal structure, flux, solid-state, phase diagram, and crystal growth. We then followed the literature in three directions: (i) the nomenclature and structural family of the Zn-free tetragonal Ba–Y silicates; (ii) Ba–Zn–Si–O compounds with firm structural evidence; and (iii) Y–Si–O systems for which high-temperature solution growth, solid-state reaction, mechanical activation, or Czochralski processes have been reported. A local full-text materials library was queried first; metadata and experimental passages were then verified through OpenAlex, Crossref, publisher pages, DOI resolution pages, IUCr, COD, RSC main texts and supplementary information (SI), and institutional repositories. The search cutoff was 2026-09-01. A miss in a database or interface means only that that channel returned nothing; it is not evidence that the literature does not exist.

2.2 Inclusion, Exclusion, and Deduplication

A publication was included if it met any of the following conditions: it directly reports the preparation or structure of the target phase; it revises the structure, phase region, or compositional series of the tetragonal Ba–Y silicate family; or it provides traceable structural data or synthesis conditions for the structurally related Ba–Zn–Si–O or Y–Si–O systems. Review articles served only to harmonize terminology and to locate primary papers; specific experimental conditions were always taken from the primary studies. We excluded

materials related only by element or function, with no crystallographic or synthetic link; phases predicted computationally but never realized experimentally; and aggregated abstracts or product information whose sources could not be traced. Glasses and glass-ceramics were treated as process-reference systems only where they clarify crystallization boundaries or competing phases; they support no inference about the structure of the target phase.

Records were first merged by DOI; where a DOI was missing or several versions existed, they were cross-deduplicated by title, authors, year, and source, and author order, volume and pages, and publication status were checked. The search returned 63 candidate records, of which 51 independent publications survived deduplication and source verification. Because the target phase has only one direct primary paper, we claim neither comprehensive coverage nor a completed independent reproduction.

2.3 Relevance Classification and Condition Extraction

We sorted the literature into five classes by how the subject of each study relates to the target phase: direct reports of the target compound; Zn-free members of the same tetragonal Ba–Y silicate family; structurally related compounds with a definite crystallographic relationship; synthesis studies usable as process references; and unrelated sources that cannot support any structural or recipe judgment. The classification applies to individual claims rather than to whole papers; the structural results and the process information from a single source may support conclusions of different scope.

For comparison, we extracted from each publication the starting materials and nominal ratios, pretreatment, mode of phase formation, temperature–time program, atmosphere and whether the system was open or closed, crucible or support, flux, cooling or growth program, and products and characterization methods. Numerical values stay attached to the original experimental record they came from and are never pooled across studies. An item the original text does not state explicitly is recorded as not reported, meaning that this check did not obtain the information, not that the original paper necessarily omitted it. Overall judgments favor results on which several studies agree for the same experimental item, and single-crystal structure, bulk phase composition, measured stoichiometry, and high-temperature stability are assessed separately.

3 The Target Compound and Related Structures

3.1 Scope of the Question

Before synthesis conditions can be compared, the object of study has to be pinned down. The available sources are the original report of the Zn-containing target phase, historical studies of the Zn-free tetragonal Ba–Y silicates, and structurally related compounds that offer nothing more than a structural reference. The original report writes the target phase as $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$ and determines its structure by single-crystal X-ray diffraction [1]; the other two groups come close to the target phase in composition or crystallographic description but cannot stand in for the Zn-containing phase itself. This section checks, in turn, the original formula and its source and the revision of the historical nomenclature by modern structural work, and then states which conclusions the existing structural and phase-region evidence can support; the synthesis variables themselves are given in Section 5.

This keeps the question “which phase was obtained” separate from the question “which conditions were used”. For each record the table retains the formula as written in the original, the typeset form used here, the structural characterization result, and the open questions. Only a study in which both the formula and direct structural evidence point to the target phase counts as direct structural evidence for it. Sources on the same tetragonal Ba–Y silicate family are used to spot misassignment, non-stoichiometry, and competing phases; structurally related compounds merely flag local structural units that need checking and do not change the definition of the target phase.

3.2 Provenance of the Target Phase

Among the publications checked here, $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$ has a single direct report [1]. In that work the crystals were grown from a high-temperature solution in an open system and the structure was identified by single-crystal X-ray diffraction [1]. It remains the only study that confirms the target phase directly, and its results cannot be merged with the conditions of related compounds into a “consensus recipe”. The report supports the target formula, the route type, and the single-crystal structure assignment, but it does not demonstrate cross-laboratory reproduction or a bulk phase-pure window, and it leaves the oxygen-stoichiometry expression, the Zn site, and the relationship to the Zn-free tetragonal silicate family only partly resolved.

Table 1 lists the original spelling and the typeset form used here in separate columns. The typeset form only harmonizes symbols, subscripts, and brackets; it neither rewrites variable-composition formulas as fixed-composition ones nor alters a published formula on the grounds of element balance.

3.3 Reassessment of the Tetragonal Silicate Family

That a historical name is still in use does not mean the structure has carried over. An early flux-growth study recorded $\text{Ba}_{5+x}\text{Y}_{13}\text{Si}_8\text{O}_{41}$ as a tentative accompanying phase [25], and a later systematic flux study folded it into a multivariable phase-formation screen, but the subject remained the Zn-free tetragonal Ba–Y silicate [42]. Modern single-crystal work then recast the related nominal composition as the non-stoichiometric modulated structure $\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ [43], and the companion ceramic study showed that bulk samples must also be checked for secondary phases and for SiO_2 picked up from the grinding medium [44]. Historical names therefore have to be re-verified against structural models and bulk phase composition; none of these studies directly observed the exact Zn-containing target formula.

An independent phase-region study combining EDS, cRED, synchrotron PXRD, and neutron refinement confirmed tetragonal $\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$ [18], giving the Zn-free Ba–Y silicates a structural reference that is independent of the tentative nomenclature. The single-crystal structure of $\text{BaY}_{16}\text{Si}_4\text{O}_{33}$ provides a Ba–Y reference structure built from isolated orthosilicate units [30]; it can be used to check how the polyhedra connect to the silicate groups, but it is not direct evidence for the Zn-containing target phase. On this basis we fix the formula of the target phase; the remaining records describe related structures and competing phases, and the scope of the claims they can support is set out in Section 4.

4 Literature Relevance and the Limits of Comparison

4.1 Basis for Classification

We organize the literature by how the subject of each study relates to the target phase rather than by assigning whole papers to a single bin. One study may supply evidence separately on structure, on synthesis route, or on characterization method, and each conclusion applies only to the object that study actually observed. The literature falls into five classes: direct reports of the target compound; Zn-free members of the same tetragonal Ba–Y silicate family; structurally related compounds with a definite crystallographic relationship; synthesis studies usable as process references; and unrelated sources that cannot support any judgment about the target structure or recipe. The classification describes how far each conclusion reaches; it is not a judgment of paper quality.

■ **Direct reports of the target compound.** The formula must match $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$, and the primary study must give both the preparation and the structure determination. At present only the open-system high-temperature solution growth and single-crystal X-ray study of Ababaikeri et al. qualifies [1]. It provides the sole direct experimental starting point, but it has not established an independently reproducible bulk phase-pure window.

■ **Same-family tetragonal Ba–Y silicates.** These studies contain no Zn but bear directly on the historical nomenclature, the structural revision, or the phase-region relationships of the tetragonal Ba–Y silicates. Early

Table 1. Compositional and structural evidence for the target compound and related Ba–Y silicates. Records are grouped by how the subject of each publication relates to the target phase; neither the Zn-free studies nor the structurally related studies constitute an independent reproduction of the Zn-containing target phase.

Formula (original → as typeset here)	Ref.	Main structural evidence	Relation to target phase	Established and open
$Ba_5Y_{12}Zn[O(SiO_4)]_8$	[1]	Single-crystal X-ray structure determination [1]	Direct report of the target compound	Established: Zn-containing target phase and its route type; open: independent reproduction, bulk phase-pure window, correspondence to the Zn-free family
$Ba_{5+x}Y_{13}Si_8O_{41}$ (tentative; x kept variable here)	[25, 42]	Reported as an accompanying phase; systematic phase-formation screening [25, 42]	Same-family tetragonal Ba–Y silicate	Established: historical name and composition-screening results; open: exact stoichiometry, structural model, relation to the target phase
$Ba_xY_{26}Si_{16}O_{71+x}$ (non-stoichiometric formula kept here)	[43]	Single-crystal revision to a non-stoichiometric modulated structure [43]	Same-family tetragonal Ba–Y silicate	Established: current structural reading of the historical formula; open: whether Zn can enter this family
$Ba_xY_{26}Si_{16}O_{71+x}$ ceramics	[44]	Bulk phase composition and microstructure [44]	Same-family tetragonal Ba–Y silicate	Established: bulk characterization requirements for the non-stoichiometric family; open: separating secondary phases and extraneous SiO_2 from the intrinsic phase region
$Ba_5Y_{13}[SiO_4]_8O_{8.5}$	[18]	EDS, cRED, synchrotron PXRD, and neutron refinement [18]	Same-family tetragonal Ba–Y silicate	Established: independently confirmed tetragonal Ba–Y structure and phase-region reference; open: site correspondence to the Zn-containing target phase
$BaY_{16}Si_4O_{33}$	[30]	Single-crystal structure with isolated orthosilicate units [30]	Structurally related compound	Established: structural reference for Ba–Y orthosilicates; open: no evidence that it is the same phase as the Zn-containing target

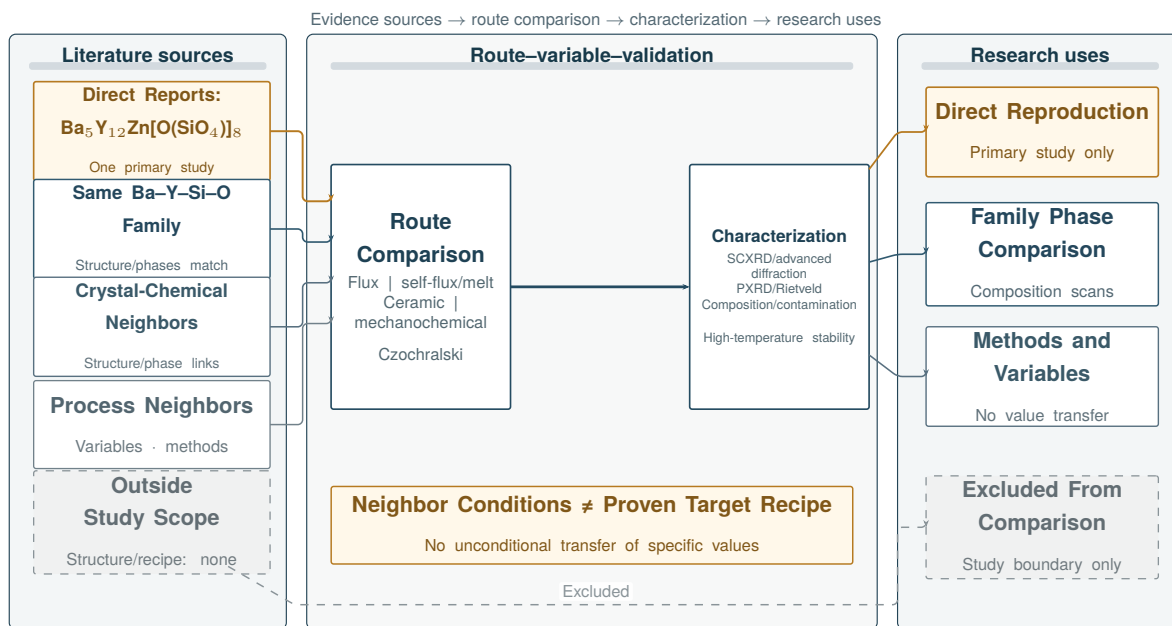


Figure 2. What each class of literature can say about synthesizing the target compound. The left column lists, in order, direct reports of the target compound, same-family tetragonal Ba–Y silicates, structurally related compounds, process-reference studies, and unrelated sources; the middle column arranges starting materials, thermal history, atmosphere, container, cooling, and product characterization under the same experimental items; the right column states the questions each class can settle. Only one direct report of the target compound exists at present [1]. Values from related systems illustrate experimental variables and characterization methods only and cannot be taken as a verified recipe for the target phase.

work reported the tentative accompanying phase $\text{Ba}_{5+x}\text{Y}_{13}\text{Si}_8\text{O}_{41}$ [25]; modern single-crystal work revised the related composition to a non-stoichiometric modulated structure [43]; and an independent study confirmed tetragonal $\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$ by several diffraction and compositional methods [18]. These results constrain the historical formulas, the composition range, and the competing phases, but they are no substitute for reproducing the Zn-containing target phase.

■ **Structurally related compounds, process references, and unrelated sources.** Ba–Zn–Si–O or Ba–Y–Si–O compounds with reliable crystallographic data can be used to compare polyhedral connectivity, cation sites, and the characterization of phase transitions [28–30, 51]; studies that supply only a synthesis route, atmosphere, cooling program, or mechanical treatment can be used to decide what to record and how to characterize the products [31, 33, 41]. None of these sources lets the composition range of the target phase be inferred. Studies that offer nothing beyond elemental similarity, an application context, or general theoretical discussion, with no crystallographic or synthetic link, are left out of the condition comparison for the target phase [8, 12, 17].

Figure 2 maps the subject of each publication to its experimental conditions and to the questions it can settle. A comparison first confirms which phase a publication actually studied and then sets starting materials, thermal history, atmosphere, container, and cooling conditions side by side under the same experimental items. This arrangement makes comparison easier but does not erase the structural differences between compounds. The more distant a publication is from the target phase, the more what can be borrowed shrinks to variable choices, experimental design, and characterization methods; specific temperatures, ratios, or flux amounts cannot be carried over directly.

4.2 Structural Relatedness

Structural relatedness requires, at a minimum, traceable crystallographic evidence; shared elements alone are not enough. $BaMSiO_4$ ($M = Co, Zn, Mg$) has been assigned by single-crystal X-ray diffraction and powder neutron diffraction to the family of stuffed-tridymite derivatives [29]. That result allows the tetrahedral networks of Zn, Mg, and Co to be compared, but the family contains no Y and belongs to a different structural series, so it is not direct evidence for the target phase. The low- and high-temperature forms of $BaZn_2Si_2O_7$ have been distinguished by neutron and X-ray diffraction [28]; the monoclinic structure of $Ba_2ZnSi_2O_7$ was established by single-crystal refinement [20]; and diffraction on $BaZnSi_3O_8$ gave the connectivity network of its ZnO_4 and SiO_4 units [51]. These studies help define which structural parameters to compare, but they cannot say under which conditions the target phase can be reproduced.

The complexity of the four-coordinated Si networks in the Ba–Si–O system, and their response to temperature, serve only as an indirect structural reference [16]. The polymorphs of $Y_2Si_2O_7$ and their thermal-expansion data show that one composition can correspond to several structures, so comparisons spanning decades must check both the structural model and the observation temperature [5, 11]. These results help frame the structural characterization questions, but they say nothing about the stability range of the target phase.

In short, same-family Ba–Y silicate studies can be used to test composition and phase region, structurally related studies to compare sites and characterization methods, and process-reference studies to sharpen experimental design. Specific synthesis conditions can be read only within the original object of study and its experimental range; they cannot be extrapolated simply because two compounds look “similar”.

4.3 What Counts as Unrelated

Containing Ba and Si is not, by itself, structural relatedness. The structure refinement of benitoite-type $BaSi_4O_9$ [14], the trigonal structure of high-pressure $BaSi_4O_9$ [19], the structure refinement of tribarium silicate [40], and the structure report on high-temperature $Ba_2[Si_4O_{10}]$ [21] all show that Ba–Si–O networks can adopt a range of structures, but none has the Y/Zn site correspondence the target phase requires. The report of 9R/6H perovskite-type structures in high-pressure Ba silicates further shows that a pressure route can lead to a different class of network altogether, from which the ambient-pressure route to the target cannot be inferred [46].

Nor can a similar application context stand in for structural evidence. Optical studies of Ce^{3+} -doped $Ba_5Si_8O_{21}$ say nothing about the identity of the Y/Zn target phase [49], and non-centrosymmetric $LiBa_2GaSi_2O_8$ resembles the target only in its optical application context, differing in both composition and structural series [17]. The classic concept of stuffed derivatives of the silica structures and the theory–experiment review of zero-thermal-expansion materials help to harmonize terminology and to understand thermomechanical behavior, but they cannot support recipe judgments for the target phase [8, 12].

Unrelated sources therefore serve only to mark the boundary of the study and do not enter the comparison of synthesis conditions for the target phase. Even where related sources appear in the same table, that placement means only that the experimental items are comparable, not that different compounds share a structure or phase-formation conditions.

5 Synthesis methods and experimental conditions compared

This section first summarizes what earlier experiments have established and then compares the synthesis methods according to the phase-formation process that dominates each. The tables list, item by item, the starting materials and ratios, thermal history, atmosphere and system, container and flux, cooling or growth program, and product characterization. The aim is not to assemble a recommended recipe from these entries but to keep the differences between the systems studied in view. The direct literature on the target phase establishes only an open-system high-temperature solution method and a single-crystal X-ray structure de-

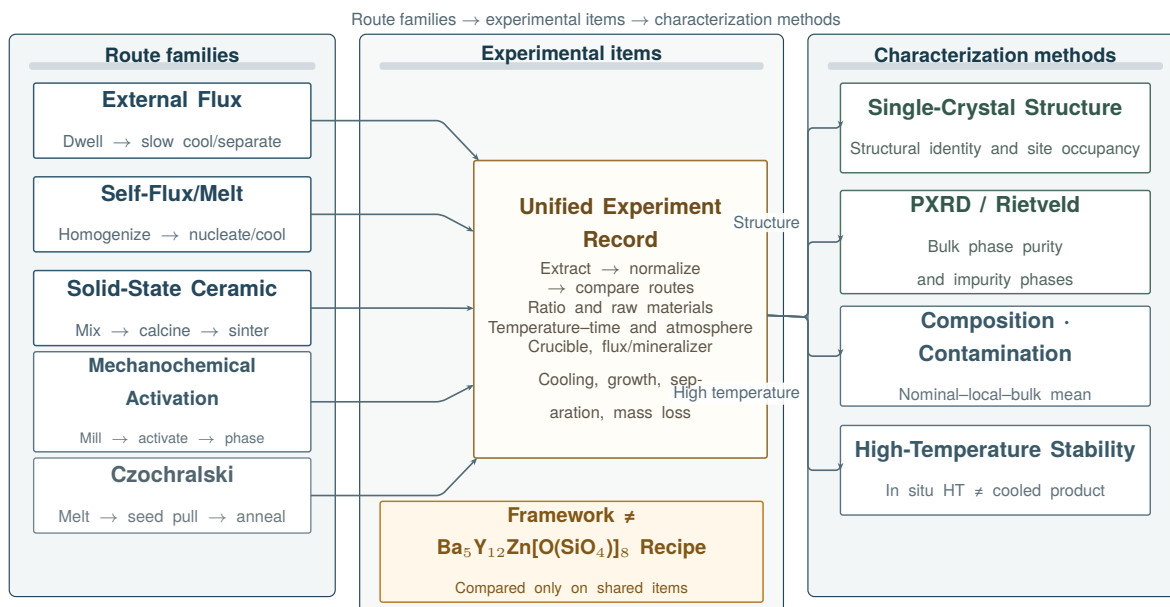


Figure 3. Conditions and product characterization compared across synthesis methods. Left: the external-flux, self-flux/melt, solid-state ceramic, mechanochemical pre-activation and Czochralski routes. Center: the comparable starting-material, thermal-history, atmosphere, container, cooling and growth conditions. Right: complementary characterization, including single-crystal structure, PXRD/Rietveld refinement, compositional analysis and high-temperature stability. The connecting lines mark the correspondence between experimental conditions and characterization results, not a causal link between a variable and phase formation; the Czochralski route is included only as a process reference, and the figure is not an experimental recipe for the target phase.

termination; the charge, temperature program, crucible, flux and cooling program are not explicitly recorded and are entered as not reported [1].

5.1 What earlier experiments established

Three conclusions follow from the existing experiments. First, composition coordinates account for the differences in phase formation in the Ba–Y–Si–O system better than a single maximum temperature does: the $\text{Y}_2\text{O}_3/\text{SiO}_2$ ratio, the amount of MoO_3 and the cooling rate together influence the structure type and the crystal quality [25, 42]. Second, a single-crystal structure does not establish bulk phase purity; the Zn-free ceramic studies still detected secondary phases [18, 44]. Third, a local phase diagram turns a reproduction problem into a testable phase-region problem and gives a basis for the next round of charges [18]. With this in mind, the experimental conditions of each route are compared below without rewriting the results of these related systems as a recipe for the target phase.

5.2 Synthesis conditions on a common footing

Figure 3 organizes the literature records into three tiers: synthesis method, experimental conditions and product characterization. For the external-flux and self-flux/melt routes it lists the flux proportion, container contact, cooling and separation, melt homogenization and nucleation conditions; for the solid-state ceramic route, the mixing, pelletizing, calcination, regrinding and refiring conditions; the mechanochemical route appears only where the source explicitly reports mechanical activation; and the Czochralski route adds the seed, pulling, rotation and annealing conditions. The connecting lines mark which experimental conditions correspond to which characterization results; they do not imply that any one variable necessarily produces a particular phase.

The number of publications behind a method says nothing about how completely its conditions are recorded. Routes such as crucible-free melt growth, open-system high-temperature solution growth or high-temperature crystallization of rare-earth silicates can be confirmed only where the source states them explicitly [2, 10, 15, 27]. A second, title-level record by Leonyuk confirms only the single-crystal object and the structure refinement; the growth method cannot be inferred from it [26]. The solid-state reaction literature spans structural phase transitions, ceramic sintering, dopant characterization and glass crystallization, but only items that can be traced to an experimental passage enter the numerical tables [9, 22, 23, 28, 32, 35, 48, 51]. Among the mechanochemical sources, Tzvetkov and Minkova explicitly report mechanical treatment, whereas Yıldırım reports only additive-assisted solid–liquid-phase sintering, which does not become mechanochemical simply because the material studied is a powder [41, 45]. The Czochralski sources likewise serve only as process references for $\text{Y}_2\text{SiO}_5/\text{Ln}_2\text{SiO}_5$ [4, 7, 31, 47].

5.3 Condition summary tables

Table 2 gives each record in one row: its literature source, chemical input, synthesis method and relation to the target phase. Table 3 continues under the same numbering with the thermal history, atmosphere and system, crucible and flux, cooling or growth program, product characterization and items not recorded. Where the source column says “experimental section”, the original experimental passage was checked; where it says “abstract” or “title”, only the object or method stated there is confirmed, and steps that are not written out are not inferred from generic terms such as “crystal growth”. A dash (“—”), or “not reported” within a cell, means that this check found no explicit record of the item, not that the original experiment lacked the step.

Table 2. Summary of synthesis conditions for the target compound and related systems (I): literature source, chemical input, synthesis method and relation to the target phase. One row per record. Zone A holds the direct literature on the target compound; zone B the Zn-free tetragonal Ba–Y silicates of the same family, which are not reproductions of the target phase; and zone C structurally related compounds and process-reference studies. “—” and “not reported” mean that no explicit record was obtained in this check, not that the original experiment lacked the step; values for related systems are not validated recipes for the target phase.

No.	Source and location	Target/product formula	Starting materials, ratio and pretreatment	Method; relation to target phase
Zone A: direct literature on the target compound				
A1	Publisher abstract; ESI used only to verify structure/composition [1]	$\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$	—	High-temperature solution; external/self-flux unclear; direct report of the target compound
Zone B: Zn-free tetragonal Ba–Y silicates of the same family; not reproductions of the target phase				
B1	p. 422, experimental section (accompanying-phase batch) [25]	$\text{Ba}_{5+x}\text{Y}_{13}\text{Si}_8\text{O}_{41}$ (accompanying phase)	Whole batch: 1 g each of $\text{BaCO}_3/\text{K}_2\text{CO}_3/\text{MoO}_3$, 0.1614 g Y_2O_3 , 0.1718 g SiO_2 ; no stoichiometric ratio for the accompanying phase can be derived	External flux; same-family tetragonal Ba–Y silicate
B2	Abstract and description of the systematic study; single-batch SI not obtained [42]	$\text{Ba}_{5+x}\text{Y}_{13}\text{Si}_8\text{O}_{41}$ (one phase among 151 screening runs)	BaO – K_2O – Y_2O_3 – SiO_2 – MoO_3 system; reagent form, single-batch ratio and pretreatment not reported	External flux; same-family tetragonal Ba–Y silicate
B3	Process excerpt from the original (self-flux crystals) [43]	$\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ ($x \approx 10.2$)	—	Self-flux/melt; same-family tetragonal Ba–Y silicate
B4	Abstract-level evidence ($9 \leq x \leq 11$ ceramic series) [43]	$\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ (series)	—	Solid-state ceramic; same-family tetragonal Ba–Y silicate

Continued on next page

Table 2. Summary of synthesis conditions (I), continued

No.	Source and location	Target/product formula	Starting materials, ratio and pretreatment	Method; relation to target phase
B5	p. 598, §2 [44]	$\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ ($9 \leq x \leq 14$ series)	$\text{BaCO}_3/\text{Y}_2\text{O}_3/\text{SiO}_2$, Ba:Y:Si= x :26:16 ($9.0 \leq x \leq 14.0$); starting materials heat-treated beforehand; dry-milled in an agate jar/balls at 400 rpm for 1 h, re-milled for 1 h after calcination	Solid-state ceramic; same-family tetragonal Ba–Y silicate
B6	Main text §2.1.1; SI §1.1/§1.3 [18]	$\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$ (phase A)	$\text{BaCO}_3/\text{Y}_2\text{O}_3/\text{SiO}_2$, $\text{BaCO}_3:\text{YO}_{1.5}:\text{SiO}_2 = 5:13:8$, final charge 0.5 g; dried/preheated, ground in agate, pressed into 10 mm pellets	Solid-state ceramic/powder; same-family tetragonal Ba–Y silicate
Zone C: structurally related and process-reference studies — high-temperature solution and melt				
C1	Publisher abstract [2]	$\text{Ba}_3\text{Zn}_4\text{Si}_4\text{O}_{15}$	—	High-temperature solution; external/self-flux unclear; structurally related compound
C2	Experimental section of the IUCr original [33]	$\beta\text{-Y}_2\text{Si}_2\text{O}_7$, with $\text{Na}_2\text{Si}_2\text{O}_5$ and oxyapatite	$\text{Na}_2\text{CO}_3/\text{Y}_2\text{O}_3/\text{SiO}_2$ in NaYSi_2O_6 proportions; nutrient: $\text{Na}_2\text{MoO}_4 = 1:10$; pretreatment not reported	External flux; process reference
C3	Record in the original paper [15]	Zn_2SiO_4	Melt of $5\text{Zn}_2\text{SiO}_4 + 3\text{Pb}_2\text{ZnSi}_2\text{O}_7$; trials with a fluoride as the Zn source; pretreatment not reported	External flux; process reference
C4	Title/metadata [10]	High-temperature rare-earth pyrosilicates	—	Crucible-free melt; process reference
C5	Indexed experimental passage of the original (flux branch) [27]	$\text{Y}_2\text{SiO}_5/\text{Y}_2\text{Si}_2\text{O}_7/\text{LaBSiO}_5$	Y–Si–O nutrient; Li/K di- and trimolybdates; exact ratio and pretreatment not reported	External flux; process reference
C6	Title and DOI metadata [26]	$\text{Y}_2\text{SiO}_5/\text{Y}_2\text{Si}_2\text{O}_7/\text{LaBSiO}_5$	—	Unclear (title does not identify the growth route); process reference
Zone C: structurally related and process-reference studies — solid-state ceramic, first group				
C7	Experimental section of the original [28]	$\text{BaZn}_2\text{Si}_2\text{O}_7$	$\text{BaCO}_3/\text{ZnO}/\text{SiO}_2$, stoichiometric; thoroughly mixed, with intermediate grinding	Solid-state ceramic; structurally related compound
C8	Experimental section of the original PDF [51]	$\text{BaZnSi}_3\text{O}_8$	BaCO_3 (99.8%), ZnO/SiO_2 (99.5%); ratio not reported; ball-milled for 5 h in a polyethylene jar/ ZrO_2 balls/deionized water, dried at 85 °C, pressed at 150 MPa	Solid-state ceramic; structurally related compound
C9	Experimental section of the original; batch-to-sample mapping not disclosed [23]	$\text{Ba}_2\text{ZnSi}_2\text{O}_7/\text{BaZnSiO}_4$; $\text{BaZn}_{2-x}\text{Mg}_x\text{Si}_2\text{O}_7$ series	$\text{SiO}_2/\text{BaCO}_3/\text{ZnO}$; Mg from a carbonate–hydroxide hydrate; ball-milled; single-batch ratio/milling parameters not reported	Solid-state ceramic; process reference
C10	Title/metadata [9]	$\text{BaZnSi}_3\text{O}_8$ -based ceramics	—	Solid-state ceramic; process reference
C11	Title/metadata [48]	$\text{Ba}_2\text{MgSi}_2\text{O}_7$ ceramics	—	Solid-state ceramic; process reference
Zone C: structurally related and process-reference studies — solid-state ceramic, second group				
C12	Experimental section of the open-access original [35]	$\text{Ba}_2\text{ZnSi}_2\text{O}_7:\text{Sm}^{3+}$	$\text{BaCO}_3/\text{ZnO}/\text{SiO}_2$ (99%), Sm_2O_3 (99.9%); nominal dopant ratio not reported; ground for 1 h in an agate mortar under ethanol	Solid-state ceramic; process reference
C13	Original abstract [32]	Ba/Zn orthosilicates: $\text{Eu}^{3+}/\text{Mn}^{2+}$	Oxides/carbonates, or $\text{Ba}_2\text{SiO}_4:\text{Eu}$ and $\text{Zn}_2\text{SiO}_4:\text{Mn}$ precursors; ratio and pretreatment not reported	Solid-state reaction; process reference

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Table 2. Summary of synthesis conditions (I), continued

No.	Source and location	Target/product formula	Starting materials, ratio and pretreatment	Method; relation to target phase
C14	Thesis metadata/ process index [22]	Ba–Zn–Si crystalline phases/ crystallized phases of high-temperature glass solders	—	Solid-state ceramic (glass-crystallization context); process reference
C15	Institutional repository abstract [45]	Y_2SiO_5 powder	$LiYO_2$ additive; ratio and pretreatment not reported	Solid–liquid-phase sintering (not evidence of mechanochemical processing); process reference
C16	Indexed experimental section of the original [50]	$Ba_2ZnSi_2O_7$	$BaCO_3$ (99.8%), ZnO/SiO_2 (99.5%); ratio not reported; ball-milled for 5 h in a polyethylene jar/ ZrO_2 balls/deionized water	Solid-state ceramic; process reference
Zone C: mechanical activation and Czochralski process references				
C17	p. 126, §2 [41]	$Y_2SiO_5/Y_2Si_2O_7$ /yttrium oxyapatite, mixed phases	$Y_2O_3/SiO_2 = 7:9$; hydrated yttria also compared; 80 cm ³ agate jar, 20 mm agate balls/40 g total, ball-to-powder ratio 7:1; in air at 256 rpm for 1–3 h	Mechanochemical pre-activation followed by heat treatment; process reference
C18	Experimental section of the original [31]	Y_2SiO_5	Y_2O_3 (99.999%)/ SiO_2 (99.99%), about 980 g, stoichiometric; mixed and ground in a corundum mortar, hydraulically pressed into blocks and prefired	Czochralski; process reference
C19	Indexed experimental passage of the original (Czochralski branch) [27]	Y_2SiO_5 (including a Cr-doped series)	High-purity Y_2O_3/SiO_2 ; exact ratio and prefiring not reported	Czochralski; process reference
C20	Index of the growth chapter of the thesis [4]	Rare-earth-doped Y_2SiO_5	—	Czochralski (chapter-level index); process reference
C21	Indexed experimental passage of the original [47]	$Y_2SiO_5/Eu:Y_2SiO_5$	Starting materials of $\geq 99.99\%$ purity; melt offset from the stoichiometric composition by at most 0.2% to compensate for SiO_2 volatilization (direction not clear from the original wording); prefiring not reported	Czochralski; process reference
C22	Abstract of the original route study [7]	Ln_2SiO_5 (including Y_2SiO_5)	—	Czochralski; process reference

Table 3. Summary of synthesis conditions (II): thermal history, atmosphere and system, crucible and flux, product characterization and items not recorded. Numbering follows Table 2; the labeled sub-items within a cell (cooling/growth, crucible/flux), separated by semicolons, carry over the corresponding columns of the original table.

No.	Temperature and time; cooling/growth	Atmosphere/system; crucible/flux	Product and characterization	Items not recorded
Zone A: direct literature on the target compound				
A1	—	Open system; gas composition not reported	Single crystals; SCXRD; yield, size and bulk phase purity not reported	Starting materials, ratio, pretreatment, temperature and time, gas, crucible, flux, cooling/ separation, yield/size, phase fraction

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Table 3. Summary of synthesis conditions (II), continued

No.	Temperature and time; cooling/growth	Atmosphere/system; crucible/flux	Product and characterization	Items not recorded
Zone B: Zn-free tetragonal Ba–Y silicates of the same family; not reproductions of the target phase				
B1	1150 °C, 3 h; cooling/growth: 2 K h ⁻¹ to 900 °C, then slow-cooled to room temperature with the furnace switched off	Air; open/closed unclear (crucible lidded); crucible/flux: lidded Pt crucible; MoO ₃ 1 g; flux phase dissolved in distilled water	Colorless prismatic accompanying crystals; tentative formula from single-crystal refinement; phase-specific yield and bulk phase fraction not reported	Stoichiometric ratio/pretreatment specific to the accompanying phase, open/closed system, phase-specific yield, phase fraction
B2	—	—	Multiphase screening; SCXRD/PXRD/Rietveld/SEM; single-batch yield of this phase not reported	Single-batch ratio, pretreatment, temperature and time, atmosphere, crucible, flux amount, cooling, yield
B3	1600 °C; time not reported	—	Self-flux single crystals; SCXRD; yield, size and bulk phase fraction not reported	Starting materials, ratio, pretreatment, time, system, crucible, flux, cooling, yield/size, phase fraction
B4	—	—	Ceramic composition window; respective roles of SCXRD/PXRD and phase fraction not reported	Starting materials, single-point ratio, pretreatment, entire thermal history/ system/container/cooling, phase fraction
B5	Calcined at 1300 °C for 12 h; fired at 1600 °C for 2 h; cooling/growth: furnace-cooled to room temperature after power-off	Air; open/closed not reported; crucible/flux: Pt–Rh plate; lidded alumina crucible; no external flux listed	Ceramics; PXRD; secondary phases and SiO ₂ contamination from the agate observed; quantitative phase fraction not reported	Heating and cooling rates, open/closed system, yield/phase fraction
B6	Calcined at 1273 K for 12 h; annealed at 1573 K for 24 h, with regrinding and refiring twice; neutron sample additionally homogenized at 1573 K for 12 h; cooling/growth: crushed and finely ground after cooling; no crystal growth	Air; heating and cooling at 5 K min ⁻¹ ; crucible/flux: alumina crucible; no flux listed	Powder; PXRD monitoring; EDS, cRED, synchrotron PXRD, neutron TOF; yield/quantitative phase fraction not reported	Phase-specific yield, quantitative phase fraction
Zone C: structurally related and process-reference studies — high-temperature solution and melt				
C1	—	Open to air; gas flow not reported	Single crystals; SCXRD; yield, size and phase fraction not reported	Starting materials, ratio, pretreatment, temperature and time, gas flow, crucible/flux, cooling/separation, yield/size/phase fraction
C2	Slowly heated to 1473 K (1200 °C), 24 h; cooling/growth: 2 K h ⁻¹ to 673 K (400 °C); subsequent cooling not reported	Atmosphere and open/closed not reported (crucible lidded); crucible/flux: lidded Pt crucible; Na ₂ MoO ₄ ; flux dissolved in boiling water	Colorless cube-shaped crystals with accompanying phases; SCXRD at 100/280 K; yield not reported	Pretreatment, atmosphere/system, subsequent cooling, yield
C3	1300 °C to 960 °C; dwell not reported; cooling/growth: 1 °C h ⁻¹ ; subsequent cooling not reported	Crucible/flux: crucible not reported; Pb ₂ ZnSi ₂ O ₇ /fluoride flux; separation not reported	Hexagonal prisms several millimeters long; chemical analysis/buoyancy density, formula Zn _{1.96} Si _{1.04} O _{4.04} as given in the source; yield not reported	Pretreatment, dwell, system, crucible, separation, subsequent cooling, yield
C4	—	Crucible/flux: crucible-free; flux not reported	Melt product; neutron powder diffraction/profile refinement; yield and phase fraction not reported	Starting materials, ratio, pretreatment, temperature and time, system, flux, cooling, yield/phase fraction

Continued on next page

Table 3. Summary of synthesis conditions (II), continued

No.	Temperature and time; cooling/growth	Atmosphere/system; crucible/flux	Product and characterization	Items not recorded
C5	—	Crucible/flux: Li/K molybdate flux; crucible, flux ratio and separation not reported	Spontaneously nucleated crystals; electron microprobe, XRD/structure refinement; yield not reported	Exact ratio, pretreatment, temperature and time, system, crucible, flux ratio/separation, cooling, yield
C6	—	—	Title supports only the single-crystal object and the structure refinement; yield/size not reported	Starting materials, ratio, pretreatment, synthesis method and all experimental conditions, yield/size
Zone C: structurally related and process-reference studies — solid-state ceramic, first group				
C7	1280 °C; time not reported	—	White polycrystalline powder; PXRD, neutron diffraction, GSAS/Rietveld; yield/phase fraction not reported	Time, system, crucible/flux, cooling, yield/phase fraction
C8	Calcined in air at 1000 °C for 3 h (5 °C min ⁻¹); sintered at 1090–1120 °C for 3 h; cooling/growth: 2 °C min ⁻¹ to 1000 °C, then furnace-cooled	Air; open/closed not reported	Ceramics; laboratory/synchrotron XRD with Rietveld refinement; yield/quantitative phase fraction not reported	Ratio, open/closed system, crucible/flux, yield/phase fraction
C9	Cooling/growth: ball-milled again after cooling; cooling rate not reported	—	Polycrystalline; HT-XRD, dilatometry; single-batch phase fraction not reported	Single-batch ratio/milling, temperature and time, system, crucible/flux, cooling rate, phase fraction
C10	—	—	Ceramics; structural/microwave-dielectric characterization; phase-purity method and phase fraction not reported	Starting materials, ratio, pretreatment and all experimental conditions, phase fraction
C11	—	—	Ceramics; structural/sintering/microwave-dielectric characterization; phase-purity method and phase fraction not reported	Starting materials, ratio, pretreatment and all experimental conditions, phase fraction
Zone C: structurally related and process-reference studies — solid-state ceramic, second group				
C12	1200 °C for 6 h in O ₂ ; heated at 5 °C min ⁻¹ ; cooling/growth: cooled naturally to room temperature, then ground	O ₂ ; open/closed not reported; crucible/flux: alumina crucible; flux not reported	Powder; XRD/FTIR/SEM-EDS/thermal and spectroscopic analysis; yield/phase fraction not reported	Dopant ratio, open/closed system, flux, yield/phase fraction
C13	—	—	Powder; PXRD/IR/UV-vis/luminescence; yield/phase fraction not reported	Ratio, pretreatment and all experimental conditions, yield/phase fraction
C14	—	—	Crystallized phases; HT-XRD/dilatometry characterization index; single-batch phase fraction not reported	Starting materials, ratio, pretreatment and all single-batch experimental conditions, phase fraction
C15	—	Crucible/flux: crucible not reported; LiYO ₂ additive amount not reported	Powder; optical microscopy, XRD, SEM-EDS; yield/phase fraction not reported	Ratio, pretreatment and entire thermal history/system/container/cooling, yield/phase fraction

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Table 3. Summary of synthesis conditions (II), continued

No.	Temperature and time; cooling/growth	Atmosphere/system; crucible/flux	Product and characterization	Items not recorded
C16	Calcined in air at 1100 °C for 3 h; sintered in air/ O_2/N_2 at 1200 °C or in N_2 -1 vol% H_2 at 1125 °C; sintering time not reported	Air/ $\text{O}_2/\text{N}_2/\text{N}_2$ -1 vol% H_2 ; open/closed not reported	Ceramics; XRD/XPS/microwave dielectric; BaSiO_3 observed in samples sintered under reducing atmosphere; yield/phase fraction not reported	Ratio, sintering time, open/closed system, crucible/flux, cooling, yield/phase fraction
Zone C: mechanical activation and Czochralski process references				
C17	Subsequent heat treatment in air for 2 h; the abstract summarizes phase formation as $T > 1000$ °C, and the specific set points are not fully given in this passage	Milling and heat treatment both in air; open/closed not reported; crucible/flux: agate balls/jar for milling; crucible and flux for the synthesis heat treatment not reported	Mixed-phase powder; XRD/IR/thermal analysis; phase fraction not reported	Specific temperatures of the synthesis heat treatment, open/closed system, heat-treatment crucible/flux, cooling, phase fraction
C18	Single-phase charge prepared at 1200 °C for 24 h; melting temperature/growth time not reported	High-purity N_2 ; open/closed not reported; crucible/flux: Ir crucible, 80×60 mm; flux not applicable	Single crystal; orientation, optical microscopy, SEM, absorption spectra; structural phase purity/yield not reported	Melting temperature/growth time, open/closed system, pulling/rotation rate/cooling, structural phase purity/yield
C19	Cooling/growth: pulled at 2.0–2.5 mm h ⁻¹ ; rotation, cooling/annealing not reported	Crucible/flux: RF-heated Ir crucible, about 80×90 mm; flux not applicable	Large crystals/spontaneously nucleated crystals; electron microprobe, XRD/structure refinement; yield not reported	Exact ratio/prefiring, melting temperature/time, system, rotation/cooling/annealing, yield
C20	—	Crucible/flux: crucible not reported; flux not applicable	Crystals/spectroscopic and crystal-field analysis; structural phase purity, yield/size not reported	Starting materials, ratio, prefiring and all growth parameters, structural phase purity/yield/size
C21	Melting temperature and growth time not reported; post-growth anneal for 20 h	Crucible/flux: RF-heated Ir crucible; flux not applicable	Crystals about 35 mm×130 mm, 460–486 g; Ir inclusions seen under the microscope; structural phase purity/yield not reported	Prefiring, melting temperature/growth time, system, pulling/rotation/cooling, annealing temperature, structural phase purity/yield
C22	—	Crucible/flux: crucible not reported; flux not applicable	Single crystals; melt stability, dopant partitioning and facet formation studied; structural phase purity/yield not reported	Starting materials, ratio, prefiring and all growth parameters, structural phase purity/yield

5.4 Comparison by zone

The central point of zone A is that the direct literature gives no complete recipe: for the target compound, only the route category and the SCXRD result can currently be checked, and no recipe item can be back-filled from related systems [1]. Zone B shows that work on the Zn-free Ba–Y silicates of the same family already covers flux-grown accompanying phases, self-flux crystals, ceramic composition scans and multi-probe solid-state/powder phase-region studies, but none of it concerns the Zn-containing target phase; in particular, the systematic screening for which no single-batch SI was obtained cannot be compressed into a single “representative condition” [18, 25, 42–44].

The direct value of zone C is that it shows how experimental conditions differ between methods. High-temperature solution papers usually report the flux composition, container and slow-cooling/separation conditions together, whereas solid-state reaction papers more often report the mixing, calcination, regrinding and sintering conditions; when the same system is prepared under different atmospheres, volatilization losses and secondary phases must also be treated as product information, and the atmosphere cannot be regarded as an isolated “optimization variable” [33, 50, 51]. C17 reports the rotation speed, duration, agate balls/jar and ball-to-powder ratio, so its mechanical activation step can be described, but the product remains a mixed-

phase powder and cannot supply a recipe for the target phase; Czochralski studies, even where they report the charge, crucible and crystal size, still need structural and phase-purity characterization, and whether the target phase can be grown cannot be inferred from the appearance of a crystal [27, 31, 41, 47]. The values in the tables therefore hold only within their own literature records: they are not merged across publications or averaged across compounds, and “not given in the abstract” is never rewritten as “not reported in the original”.

6 Lessons from the Published Experiments

Comparing conditions side by side does more than preserve the numbers; it also brings out patterns that recur across independent studies. Taken together, the published experiments on the target phase and its relatives yield three lessons that bear directly on the next round of experiments.

6.1 Phase Selection Is Governed by Composition Coordinates, Not by a Single Maximum Temperature

Across the 151 flux-growth experiments in the Ba–Y–Si–O system, the $\text{Y}_2\text{O}_3/\text{SiO}_2$ ratio and the amount of MoO_3 control both the connectivity of the silicate anions and the type of crystal that forms. Lowering the $\text{Y}_2\text{O}_3/\text{SiO}_2$ ratio, or raising the MoO_3 content within a certain range, drives the structures toward higher SiO_4 connectivity; an excess of MoO_3 hinders crystal growth, and slower cooling generally improves crystal morphology and yield [42]. Screening for the target phase should therefore treat the nutrient composition, the flux ratio, and the cooling rate as independent variables rather than simply reproducing one maximum temperature. The early accompanying crystals were obtained by soaking at 1150 °C followed by slow cooling, but they came from a whole-batch $\text{K}_2\text{CO}_3/\text{MoO}_3$ flux system, and that result cannot be converted into a stoichiometric starting composition for the target phase [25].

Solid-state work points the same way. In the $\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ ceramic series with $9 \leq x \leq 14$, the main and secondary phases shift with the nominal Ba content, and SiO_2 picked up from agate grinding blurs the identification of the intrinsic phase region [44]. $\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$ was located by a powder composition screen that traversed the phase diagram and was then confirmed by repeated annealing and regrinding together with several diffraction and compositional-analysis techniques [18]. What the target phase lacks, then, is not one more isolated temperature but a local phase diagram of the Ba–Y–Zn–Si–O system, with quantitative phase analysis at every composition coordinate.

6.2 Single-Crystal Evidence Versus the Bulk Phase-Purity Window

The direct report on the target phase confirmed its structure by single-crystal diffraction, but the available material does not also supply an independently reproducible starting composition, thermal history, or bulk phase fraction [1]. Ceramic studies of the Zn-free series show that even after the main phase has formed, a bulk sample may still carry secondary phases and contamination from the processing media [18, 44]. A reproduction experiment therefore needs at least two complementary kinds of characterization: single-crystal or advanced diffraction to confirm the structural model, and quantitative PXRD/Rietveld analysis to measure the target-phase fraction in the whole batch. Without both, “obtaining target crystals” is not the same as “establishing a phase-pure preparation method”.

6.3 The Local Phase Diagram as the Central Experiment Linking Reproduction to Discovery

The discovery of $\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$ shows that even in a phase region as well populated as BaO– Y_2O_3 – SiO_2 , a new phase can still sit at the boundary with competing phases or with a glass-forming region. The investigators closed in on the target region only gradually, updating each new batch of starting compositions from the overall composition, the known phase proportions, and compositional analysis [18]. The Zn-containing system needs this strategy all the more, because adding Zn changes the cation count, the

oxygen stoichiometry, and possibly the liquid-phase behavior all at once. Phase-diagram experiments can distinguish three cases: a line compound of narrow composition, a limited solid-solution range within the Zn-free series, and a metastable crystalline phase that forms only along a particular liquid-phase path. Any of these three outcomes carries more structural information than repeated trials at a single temperature.

7 Synthesis Routes to Related Compounds

This section compares the synthesis routes to related compounds by stage of phase formation rather than paper by paper. Figure 3 summarizes the experimental items that can be compared, and Table 2 keeps the numerical values together with their sources. The discussion is limited to the methodological information these studies provide; no route is recast as a candidate recipe for $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$.

7.1 High-Temperature Solution Routes

The Y–Si–O literature makes a first point: “obtaining crystals” is not the same as “assigning the structure correctly”. The $\beta\text{-Y}_2\text{Si}_2\text{O}_7$ grown from a Na_2MoO_4 flux came with other silicate crystals, and the final phase assignment rested on single-crystal structure determination; studies of this kind offer reference points for the nutrient-to-flux ratio, the use of a lidded crucible, slow cooling, and crystal separation, but they cannot establish the stability range of the target phase [33]. A re-examination of the $\text{Y}_2\text{Si}_2\text{O}_7/\text{Y}_2\text{SiO}_5$ polymorphs by ^{89}Y MAS-NMR shows that a local probe can complement diffraction in telling polymorphs apart, but cannot on its own prove that the target phase has formed [6]. The review of Y–Si–O synthesis and the classic crystal-chemical classification of rare-earth silicates can standardize the terminology for polymorphs and the degree of silicate polymerization, but they are no substitute for the original experimental data [13, 34].

In the Ba–Zn–Si–O system, the open-system high-temperature solution study established only the structure of $\text{Ba}_3\text{Zn}_4\text{Si}_4\text{O}_{15}$ and the class of route used; the exact batch composition and thermal history are still recorded as not reported [2]. The early molten-solution growth of Zn_2SiO_4 shows that the slow-cooling schedule, the method of crystal separation, and the flux composition have to be reported together; conditions involving Pb/F also call for review of toxicity, volatilization, and waste disposal and cannot be a default choice [15]. The clearly sourced crucible-free melt study and the high-temperature crystallization study of $\text{Y}_2\text{SiO}_5/\text{Y}_2\text{Si}_2\text{O}_7$ offer reference points for container contact, spontaneous nucleation, and structural identification, but they cannot show that the target phase behaves the same way in a melt [10, 27]. The second Leonyuk paper is supported by title-level evidence only; it confirms its single-crystal subject and structure refinement, and nothing about the synthesis route can be inferred from it [26].

■ **Implications for the target-phase experiments.** List the starting composition and the flux composition separately, note the container contact, and record the cooling and separation steps in full; crystal-structure determination, bulk diffraction, and compositional analysis should be run as a matched set. None of the following transfers directly: the specific values reported for related compounds, the suitability of Pb/F systems, or an assignment of the target phase from crystal habit.

7.2 Solid-State Reaction Routes

For BaMSiO_4 ($\text{M} = \text{Co}, \text{Zn}, \text{Mg}$), single-crystal X-ray and powder neutron diffraction established an isostructural stuffed-tridymite series, showing that isovalent substitution on the tetrahedral site can preserve the parent topology; that framework, however, is not equivalent to the zero-dimensional mixed building units of the target phase [29]. The single-crystal refinement of monoclinic $\text{Ba}_2\text{ZnSi}_2\text{O}_7$ and the structure report on monoclinic $\text{Ba}_2\text{MgSi}_2\text{O}_7$ supply the Zn and Mg references, respectively; both help in identifying competing phases and checking substitution risks, but neither supports a claim that they are isostructural with the target phase [3, 20].

Controlled substitution series show how composition, structure, thermal history, and characterization should be reported as a matched set. The Ba/Sr-substituted compounds were confirmed by SCXRD together

with EDS, and HT-XRD on $BaZn_{2-x}Co_xSi_2O_7$ and on the Ba/Sr–Zn–Si/Ge series mapped composition-dependent regions of structure retention and phase boundaries. These studies make clear that the nominal degree of substitution must be tied to the measured composition and the in situ structural results, and that the Y/Zn tolerance of the target phase cannot be quantified by transfer from another system [24, 36, 38]. The review of $BaZn_2Si_2O_7$ solid solutions organizes the background on how thermal expansion varies with composition, but it is no substitute for an individual original sintering batch [37]. The glass-ceramic study containing $Ba_{1-x}Sr_xZn_2Si_2O_7$ likewise provides only functional and crystallization background, not a direct basis for synthesizing the target phase [39]. The controlled comparison of $Ba_2ZnSi_2O_7$ ceramics under several atmospheres shows that atmosphere effects must be read together with Zn loss, secondary phases, and surface valence states; a change in dielectric properties alone is no measure of structural stability [50].

■ **Implications for the target-phase experiments.** Report the stoichiometric weighing, the regrinding and refiring cycles, the atmosphere, and the measured composition side by side, and characterize by PXRD/Rietveld analysis, compositional analysis, and HT-XRD where needed. The isovalent-substitution, thermal-expansion, and dielectric results of related studies cannot be rewritten as the aliovalent-substitution mechanism, phase-pure range, or reduction stability of the target phase.

7.3 Mechanochemical Pre-Activation and Czochralski Growth as Process References

The mechanochemical literature supports only one conclusion: pre-activation may change the subsequent phase-formation path. The yttrium oxyapatite study specified milling at 256 rpm in air for 1–3 h in an agate jar with agate balls; heat treatment was still required afterwards, and several Y–Si–O phases coexisted. Those values describe that mixed-phase system alone and cannot be taken over as activation conditions for the target phase [41]. The Y_2SiO_5 powder study explicitly reported only $LiYO_2$ -assisted solid–liquid-phase sintering with microscopy, XRD, and SEM–EDS characterization; a powder subject does not make it a mechanochemical study [45].

Czochralski studies offer reference points for the melt, the Ir crucible, the atmosphere, SiO_2 volatilization, pulling and rotation rates, and annealing. The combined growth, morphology, and spectroscopy study of Y_2SiO_5 shows that structural and compositional identification is still needed after the crystal has been grown. The early Czochralski work on rare-earth orthosilicates and the later thesis can serve to standardize how process parameters are reported, but there is at present no evidence of Czochralski growth of the target phase [4, 7, 31, 47].

■ **Implications for the target-phase experiments.** The contamination checks used in mechanochemical activation and the control of melt, crucible, and volatilization in Czochralski growth are useful references; at the same time, the bulk phase composition, the measured element ratios, inclusions, and the single-crystal structure must be verified. Ordinary mixing does not qualify as mechanochemical activation, and the fact that Y_2SiO_5/Ln_2SiO_5 can be pulled from the melt does not mean the target phase can. All numerical values are limited to the original literature data listed in Table 2.

8 Structural and Compositional Characterization of the Products

The four kinds of characterization listed on the right of Figure 3 answer different questions and cannot stand in for one another: single-crystal or advanced diffraction confirms the structural model, PXRD/Rietveld analysis establishes the bulk phase composition, chemical analysis and microscopy check the measured element ratios and contamination, and high-temperature experiments delimit the stability range under a given atmosphere and thermal history. The “Products and characterization” column of Table 3 accordingly bounds what each experimental record can support. Where one kind of data is missing, the corresponding conclusion is simply held with caution; the other methods cannot fill the gap.

8.1 Structural Model

Single-crystal X-ray diffraction ties the identification of the target phase to a structural model of the crystal actually measured; the existing direct study identified $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$ on precisely this basis [1]. Among the related Zn-free Ba–Y silicates, single-crystal refinement revealed a non-stoichiometric modulated structure, and cRED, synchrotron PXRD, and neutron refinement together confirmed an independent tetragonal Ba–Y phase region [18, 43]. Other single-crystal and synchrotron studies established the structures of their own Ba–Y and Ba–Zn–Si compounds, but those conclusions hold only for those compounds [30, 51]. A single-crystal structure proves that the measured crystal matches the refined model; it does not by itself prove that the whole batch is phase-pure, nor does it let the nominal starting composition stand in for the measured site occupancies. Local spectroscopy at most sharpens the discrimination of polymorphs and offers no direct evidence that the target phase has formed [6].

8.2 Bulk Phase Composition

PXRD/Rietveld analysis addresses the bulk sample: phase fractions, unindexed reflections, and differences in polymorph under the same thermal history. The Ba–Y ceramic study detected secondary phases by powder diffraction, and the study of the independent phase region combined synchrotron powder data with other diffraction methods to confirm the phase composition [18, 44]; the Ba–Zn–Si studies likewise drew on X-ray, neutron, or synchrotron data to separate the bulk polymorph from the structural model [28, 51]. PXRD/Rietveld analysis therefore supports judgments about bulk phase composition and impurity phases, but similar peak positions or the mere appearance of the main phase never imply phase purity, and powder data cannot replace single-crystal determination of site occupancies. From studies that report only solid-state transitions or thermally induced changes, this review borrows the characterization methods alone; the word “optimized” in a title is not taken as a criterion for the target phase [9, 23].

8.3 Measured Composition and Contamination

The nominal ratio, the local spot compositions, and the overall composition of the sample have to be recorded separately. EDS, EPMA, or ICP can check for compositional differences between grains, foreign phases, and material introduced by the processing media. The Zn-free Ba–Y ceramic study brought both the secondary phases and the agate-derived SiO_2 contamination into its analysis [44]; the study of the independent Ba–Y phase region paired EDS with several diffraction methods to constrain composition and structure [18]; and the related Ba/Sr–Zn–Si compounds had their own compositional relationships confirmed by single-crystal structure and EDS in combination [38]. These methods account only for the elements detected within the analyzed spot or sampled volume: a handful of spots cannot establish the homogeneity of a whole batch, and the nominal pre-firing ratio cannot be rewritten as the post-firing chemical formula. Combined microscopic, thermal, and spectroscopic measurements can provide process references for doping or crystal growth, but they do not carry the compositional conclusions for the target phase [31, 35].

8.4 High-Temperature Stability

HT-XRD, thermal analysis, dilatometry, and multi-atmosphere experiments probe how a material changes under a given thermal history or atmosphere; they do not replace room-temperature structural identification. The low- and high-temperature phases of $\text{BaZn}_2\text{Si}_2\text{O}_7$ were distinguished by neutron and X-ray diffraction, which shows that the in situ high-temperature phase cannot be equated with the product after cooling [28]; the related Zn/Co and Ba/Sr–Zn–Si/Ge studies went further and tied the substituted composition to thermal stability or phase boundaries [24, 36]. The tolerance of a related ceramic to one particular reducing atmosphere serves only as an atmosphere reference and does not show that the target phase is equally stable [50]. High-temperature experiments can delimit the transition, decomposition, or atmosphere-tolerance range of a published system, but in situ signals alone cannot prove that the cooled sample is phase-pure; where the relevant data are missing, the stability conclusion stays undetermined.

Together, the four kinds of characterization set the scope of any conclusion: single-crystal or advanced diffraction confirms the structural model, powder diffraction establishes the bulk phase composition, chemical analysis verifies the measured element ratios and contamination, and high-temperature experiments delimit the thermal-history and atmosphere boundaries. Only when these results are matched to the synthesis conditions of the same batch can one discuss whether the target phase has been reproduced and how far the related processes can be extrapolated.

9 Common Experimental Problems and Limits of Applicability

The range over which a method applies can be judged only when failed or anomalous results are recorded together with the composition, the experimental conditions, and the product characterization. This section covers phase competition, contamination and volatilization, atmosphere and polymorphism, missing information, and environmental, health, and safety (EHS) constraints. It is confined to published conditions and does not turn these boundaries into new experimental protocols.

■ **Phase regions and phase competition.** The systematic comparison of flux growth across the Ba–Y silicate family shows that phase-formation outcomes must be read against the composition and route variables; no single heat-treatment condition can be lifted out as a general one [42]. Later work, through re-examination of the non-stoichiometric structure, ceramic composition series, and several diffraction methods, revealed modulated structures, secondary phases, and competing phases or glass-forming regions [18, 43, 44]. These results delimit only the composition ranges each study actually examined; they cannot show that no other phases exist in the regions left uncovered. Mechanochemical studies indicate only that pre-activation may influence subsequent phase formation, and a particular negative result cannot be read as an intrinsic phase boundary of the target phase [41]. The products, characterization, and synthesis conditions in Tables 2 and 3 must therefore come from the same publication, and impurity phases or glass cannot be interpreted apart from the composition and thermal history.

■ **Contamination, volatilization, and flux boundaries.** The Zn-free Ba–Y ceramic study traced the secondary phases and the agate-derived SiO_2 contamination to specific processing steps: an apparent departure from the nominal ratio should first prompt a check of the starting materials and the processing, and cannot automatically be taken as evidence of genuine solid solution [44]. The Czochralski study of the Y–Si–O system offers reference points for the melt, crucible, atmosphere, and pulling conditions, but the available material is not enough to judge whether the target phase loses Zn or Si by volatilization [47]. Lead- and fluorine-containing fluxes were once used to grow Zn–Si–O crystals, but those early conditions serve only for comparing slow cooling, separation, and the role of the flux; they are no basis for choosing materials for the target phase [15]. Pickup from the grinding media, mass loss in open systems, and flux residues or flux-driven phase selection should therefore be recorded separately; where the literature gives no pre- and post-firing compositions or residue analysis, this review assigns no specific failure mechanism.

■ **Atmosphere and polymorphism.** The low-temperature monoclinic and high-temperature orthorhombic phases of $\text{BaZn}_2\text{Si}_2\text{O}_7$ were distinguished by X-ray and neutron diffraction, so the in situ high-temperature phase and the cooled product are not interchangeable [28]. High-temperature diffraction further shows that Zn/Co substitution and the composition of the Ba/Sr–Zn–Si/Ge solid solutions are coupled to phase stability or polymorph boundaries; if composition, temperature, or the time point of observation is missing, the stability range cannot be defined [24, 36]. The reduction-resistance report on a related Ba–Zn–Si ceramic shows only that that compound was tested for atmosphere tolerance under the published conditions; it cannot be extrapolated to the reduction stability of the target phase [50]. Other comparisons of thermal expansion and high-temperature diffraction likewise serve only as references for thermally induced change [23].

■ **Missing information and EHS.** Some publications give only the synthesis method or the structural conclusion without the full set of ratios, temperature–time program, atmosphere, crucible, and cool-

ing [1, 2, 9, 20, 48]. This means only that this review has not obtained the information; it does not mean that the original paper failed to report it, nor that the step was absent from the experiment. Table 2 marks such entries with “—” and preserves that boundary. The less complete the record, the fewer specific conditions can be drawn from it. EHS boundaries are likewise described only as published: Ba-containing powders, species volatile at high temperature, the Pb/F flux conditions of the early literature, crucible compatibility, and waste streams each require separate review, and the Pb/F routes in particular must not be promoted from early examples into recommended protocols [15]. Any reuse must pass independent EHS, container-compatibility, and waste-segregation review; this review supplies no operating parameters and makes no presumption of compliance about ventilation, protective measures, equipment, or waste classification that the sources do not report.

These four kinds of boundary cannot be collapsed into a single unverified failure mechanism. Phase competition holds only within the published composition and thermal-history ranges; contamination or mass loss needs support from analysis of the media, of the pre- and post-firing compositions, or of residues; judgments about atmosphere and polymorphism must keep both the observation temperature and the state after cooling; and missing information marks only the limit of present knowledge. Accordingly, where the literature reports only “accompanied by” or gives only structural results, the text says “accompanied by” or “consistent with”, never “caused”. This keeps both negative results and records of success while avoiding any conflation of unreported items, process anomalies, and genuine phase regions.

10 Priority Research Directions and Experimental Design

The following schemes are proposed jointly from the existing evidence and the model candidates; all are experimental hypotheses rather than verified recipes for the target phase. We recommend proceeding in the order “local phase diagram—site occupancy/oxygen stoichiometry—crystal growth—analogue substitution”; the phase-composition results of each stage determine whether the next stage is worth undertaking.

10.1 Direction 1: A Local Phase Diagram near the Target Composition

The first priority is to establish a fine-step composition grid around $\text{Ba}_5\text{Y}_{12}\text{ZnSi}_8\text{O}_{40}$. The base precursors are the model’s top-ranked and chemically conventional BaCO_3 , Y_2O_3 , ZnO , and SiO_2 , with the target stoichiometry corresponding to a molar ratio of 5:6:1:8. This direction is a model prediction awaiting experimental verification. We suggest first screening in stages over the temperature range of the Zn-free solid-state studies: an initial reaction at about 1000 °C for 12 h, followed by regrinding and annealing at 1300 °C for 24 h, with repetition decided on the basis of the PXRD results; short-duration points at 1450–1600 °C are added only if obvious unreacted material remains and the mass loss and crucible compatibility are controllable. The temperatures and times are taken from the screening ranges of the neighboring Ba–Y solid-state studies [18, 44]; they serve only to organize the experimental matrix and do not imply that the target phase necessarily forms under these conditions.

The composition axes should include at least: (i) small deviations of the Ba/Y ratio around the target point; (ii) variation of the Zn content; and (iii) the Zn-free reference phase related to the oxygen stoichiometry. For every sample, the pre- and post-firing masses, the thermal history, and the cooling mode are recorded, and the phase composition and measured element ratios are given by quantitative PXRD/Rietveld analysis, SEM–EDS/EPMA, and ICP analysis where necessary. If glass or a multiphase boundary appears, sampling should be densified in that region rather than retaining only the most ideal diffraction pattern.

10.2 Direction 2: Zn–Y–Oxygen-Stoichiometry Coupling

A testable charge-balanced series is

$$\text{Ba}_5\text{Y}_{13-x}\text{Zn}_x[\text{SiO}_4]_8\text{O}_{8.5-x/2}, \quad 0 \leq x \leq 1.25,$$

where $x = 0$ corresponds to the reported Zn-free tetragonal reference phase and $x = 1$ corresponds, in elemental stoichiometry, to the target composition. We suggest $x = 0, 0.25, 0.50, 0.75, 1.00, 1.25$ as the discrete points of the first round. This direction is a model prediction awaiting experimental verification. The precursors remain BaCO_3 , Y_2O_3 , and SiO_2 , with the Zn content adjusted by ZnO; the expression is only a charge-balance hypothesis and does not presuppose the true site occupancy of Zn or the distribution of oxygen vacancies. Assessment should examine simultaneously the continuity of the lattice parameters, the phase fractions, the measured Zn/Y ratio, and the site-occupancy refinement; where feasible, neutron diffraction or independent measurements of oxygen content and valence state can further distinguish Zn substitution, oxygen defects, and a two-phase mixture.

The scientific value of this series exceeds that of a single-point reproduction: a continuous change in the structural parameters supports limited solid solution; essentially unchanged lattice parameters with a changing two-phase ratio is closer to a two-phase region; and new diffraction peaks appearing only at $x \approx 1$ suggest a line compound. All three outcomes constrain the formation mechanism of the target phase.

10.3 Direction 3: Solid-State Reaction in Parallel with High-Temperature Solution Crystal Growth

The solid-state reaction route serves to locate the bulk phase region, and the high-temperature solution route serves to obtain crystals suitable for SCXRD analysis. This direction is a model prediction awaiting experimental verification. The high-temperature solution route can take the $\text{K}_2\text{CO}_3/\text{MoO}_3$ flux system already validated in the Ba–Y–Si–O system as its reference starting point, with the BaCO_3 , Y_2O_3 , ZnO, and SiO_2 precursors of Direction 1; the maximum temperature is set within 1100–1200 °C, held for about 3 h, followed by slow cooling at 1–2 K h^{−1} to about 900 °C, while the nutrient-to-flux ratio and the $\text{Y}_2\text{O}_3/\text{SiO}_2$ ratio are varied systematically [25, 42]. These values come from crystal growth in the Zn-free homologous series; they must be compared in parallel with the solid-state reaction route using the target starting composition and cannot be called a recipe for the target phase.

Each crystal-growth batch should report at least the liquid-phase composition, the crucible filling ratio, the maximum temperature, the dwell time, the cooling rate, the separation method, the crystal size, and the yield; the crystals are verified by SCXRD and EDS, and the residual mother liquor is checked by PXRD and compositional analysis. If only a few target single crystals are obtained while the composition of the mother liquor is far from the target composition, this should be interpreted as selective growth from the liquid phase rather than as a bulk phase region.

10.4 Direction 4: Mg/Co Analogs and Model-Proposed Precursors

This review invoked a local two-step inorganic retrosynthesis model to predict Top-5 precursor combinations for three elemental formulae. The model's top-ranked combinations for the Zn, Mg, and Co targets all retain BaCO_3 , Y_2O_3 , and SiO_2 , with ZnO, MgO, and Co_3O_4 as the fourth starting material, respectively; the probabilities of the three top-ranked combinations within their respective candidate pools are 0.594, 0.522, and 0.298. This direction is a model prediction awaiting experimental verification. Table 4 converts the model output into minimal testable experiments. The model proposes precursor combinations solely on the basis of elemental stoichiometry and is not used to judge the silicate structural units; the text throughout retains the structural formula used in the target publication.

The remaining Zn routes given by the model include additional BaO, BaF_2 , or $\text{Ba}(\text{NO}_3)_2$, or non-standard database labels; their probabilities are all clearly lower than that of the top-ranked combination, and they should not be recommended alongside it. An additional Ba source would alter the local phase diagram; BaF_2 increases the risk of fluorine-containing volatiles and waste streams; and non-standard labels may further reflect normalization problems in the training data. MgCO_3 in the Mg routes can serve as a separate control for precursor form but should not be added together with MgO; the several Co oxide precursors should not be regarded as three equivalent schemes.

10.5 Organizing Experiments for Scientific Discovery

Each round of experiments should be judged by the criterion of “maximizing distinguishable conclusions” rather than aiming merely at obtaining crystals. The most informative combination is: (i) a fine-step composition grid coupled with quantitative phase analysis to establish the local phase diagram; (ii) parallel solid-state reaction and flux-growth experiments at the same composition to distinguish the equilibrium phase region from selective nucleation in the liquid phase; and (iii) a coupled Zn-content and oxygen-stoichiometry series to test the site and defect hypotheses. All negative results are likewise preserved, because they determine the composition choices of the next round and constrain the boundary of “synthesizability”.

11 Conclusions

$\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$ rests on sound single-crystal structural evidence and on evidence of synthesis by the high-temperature solution method, but no public evidence yet shows that its bulk phase-pure range has been independently reproduced [1]. At this stage it would be premature to prescribe a “best recipe”; what can be specified are the variables that must be controlled together and the characterization steps that go with them.

The neighboring systems supply three consistent lessons. First, phase formation in the Ba–Y–Si–O system is sensitive to the composition coordinates, the MoO_3 content, and the cooling path, and the Zn-free tetragonal phase additionally involves non-stoichiometry, competing phases, and a glass-forming region [18, 42, 43]. Second, secondary phases in ceramic samples change how the nominal composition must be interpreted, and a single-crystal assignment is no substitute for bulk phase quantification [44]. Third, the polymorphism and phase stability of the Ba–Zn–Si–O system vary with thermal history and substituted composition, so the Zn source, the cooling path, and the observation temperature must be recorded together [24, 28, 50]. These lessons can guide experimental design, but the numerical values of the related systems cannot be carried over as conditions for the target phase.

The most informative work for the next stage is, in order: (i) establishing a local phase diagram around the target composition from BaCO_3 , Y_2O_3 , ZnO , and SiO_2 as the base starting materials; (ii) examining the $\text{Ba}_5\text{Y}_{13-x}\text{Zn}_x[\text{SiO}_4]_8\text{O}_{8.5-x/2}$ series to distinguish limited solid solution, a line compound, and a two-phase region; (iii) running solid-state reaction and high-temperature solution growth in parallel, with the bulk phase

Table 4. Top-ranked precursor combinations from the two-step inorganic retrosynthesis model and their experimental uses. All combinations are model-ranked results (not validated by chemical routes); the process conditions are taken from the literature on neighboring systems and require re-checking by phase-diagram experiments.

Hypothetical target	Model top-ranked precursors	Candidate-pool probability	Suggested experiment and criteria
Zn target	$\text{BaCO}_3 + \text{Y}_2\text{O}_3 + \text{SiO}_2 + \text{ZnO}$	0.594	Base starting composition for the local phase diagram; solid-state screening first, then transfer of near-single-phase compositions to flux growth; bulk phase region and structural assignment determined by PXRD/Rietveld analysis and SCXRD, respectively
Mg analog	$\text{BaCO}_3 + \text{Y}_2\text{O}_3 + \text{SiO}_2 + \text{MgO}$	0.522	Mg^{2+} as an isovalent size control; screening along the same composition/thermal-history matrix as for Zn; substitution on the same site discussed only if a new phase appears and the measured composition varies continuously
Co analog	$\text{BaCO}_3 + \text{Y}_2\text{O}_3 + \text{SiO}_2 + \text{Co}_3\text{O}_4$	0.298	Air atmosphere as the first-round condition; measured composition and phase fractions determined; secondary phases not to be mistaken for structural substitution

region fixed by PXRD/Rietveld analysis and the structural assignment by SCXRD; and (iv) testing the size difference through the Mg/Co analogs only once the target phase has been independently reproduced. The oxide or carbonate combinations recommended by the retrosynthesis model can serve as a starting point for the precursors, but they cannot replace phase-diagram, thermal-history, and structural verification.

The experiment most worth starting now is a fine-step local phase diagram around the target composition: $BaCO_3$, Y_2O_3 , ZnO , and SiO_2 as precursors, staged calcination and annealing within the temperature range of the neighboring systems, and quantitative PXRD/Rietveld analysis plus EDS or EPMA at every composition point, with the first aim of establishing whether the target phase has a reproducible bulk phase region; the near-single-phase compositions are then carried into high-temperature solution growth and the structure is re-checked by SCXRD. Such experiments update the hypotheses whether they succeed or fail. A fine-step composition grid, the measured element ratios and oxygen stoichiometry, cross-validation between single-crystal and powder diffraction, and fully preserved negative results together form an iterable experimental basis and bring the following questions into focus: whether Zn enters the Zn-free tetragonal silicate series, whether the target phase has a reproducible stability region, and whether the crystal-growth conditions and the bulk phase-formation conditions belong to the same phase-equilibrium region.

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